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1

Online, Offline, and Real-Time Adaptive Radiotherapy

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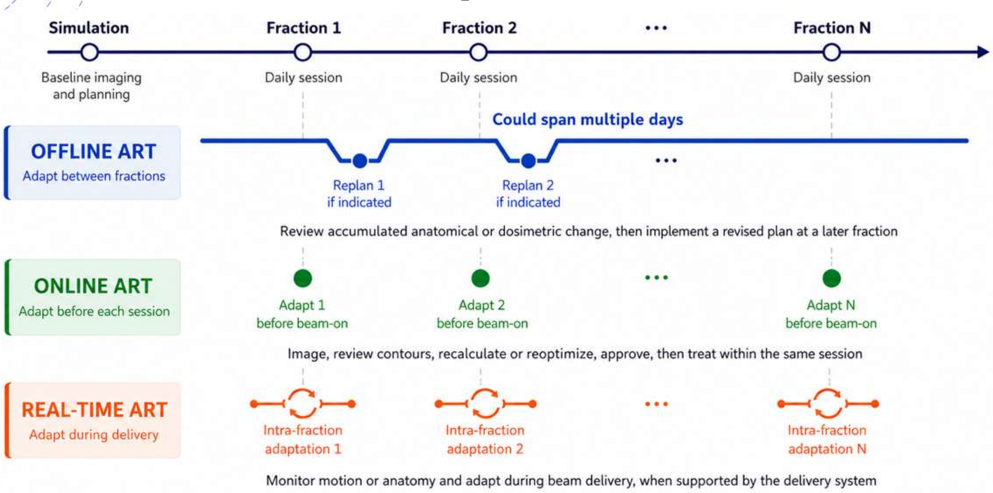
2

Outline

- Adaptive Radiotherapy in Context
  - Offline ART
  - Online ART
  - Real-Time ART
- Comparing Adaptive Approaches
- Clinical Implementation and Quality Assurance
- Future Directions and Emerging Frontiers

3

Overview of Adaptive Workflows



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4

Offline Adaptive Radiotherapy

- Offline ART: refers to adaptive replanning performed after treatment has started, but outside the treatment session.
- Common triggers
  - Tumor shrinkage or progression
  - Weight loss or body contour change
  - OAR displacement
  - Pleural effusion or atelectasis
  - Systematic setup or anatomy change
  - ...

5

Offline Adaptive Radiotherapy

- Typical workflow
  1. Review prior treatment images
  2. Identify clinically meaningful change
  3. Recalculate dose or evaluate plan adequacy
  4. Generate revised plan
  5. QA and approve new plan
  6. Implement at a later fraction

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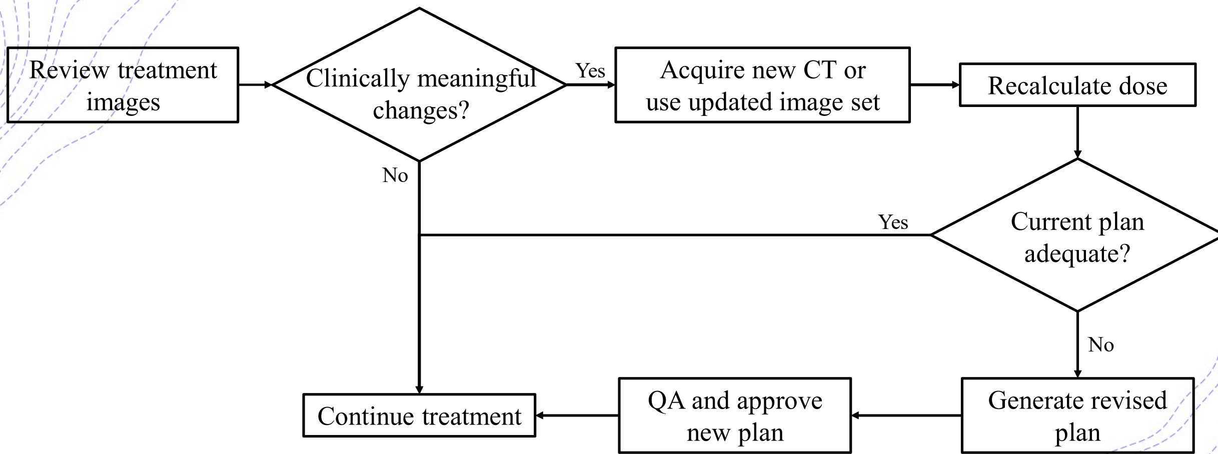
6

Offline ART: Flowchart



7

Offline ART: Strengths and Limitations

Strengths
- More time for review, planning, and QA
- Fits conventional clinical workflows
- Minimal impact on daily treatment slot
- Useful for progressive anatomical change
- Easier to implement than online ART

Limitations
- Delayed response to anatomy change
- Does not address anatomy-of-the-day
- May miss rapid or transient changes
- Requires repeat imaging and replanning resources
- Adapted plan starts at a later fraction

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8

Online Adaptive Radiotherapy

- Online ART: adapts the treatment plan after daily imaging and before treatment delivery in the same session.
- Common triggers
  - Daily target position or shape variation
  - Bladder or rectal filling changes
  - Bowel proximity to target
  - Rectal gas or bowel gas
  - OAR position changes near high-dose region
  - Poor target coverage on scheduled plan
  - OAR constraint violation on anatomy-of-the-day
  - Need for hypofractionated treatment near critical structures
  - …

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9

Online Adaptive Radiotherapy

- Typical workflow
  1. Acquire daily treatment image
  2. Register daily image to reference image
  3. Review and edit contours
  4. Evaluate scheduled plan on anatomy-of-the-day
  5. Adapt or reoptimize plan if needed
  6. QA and approve adapted plan
  7. Deliver treatment in the same session

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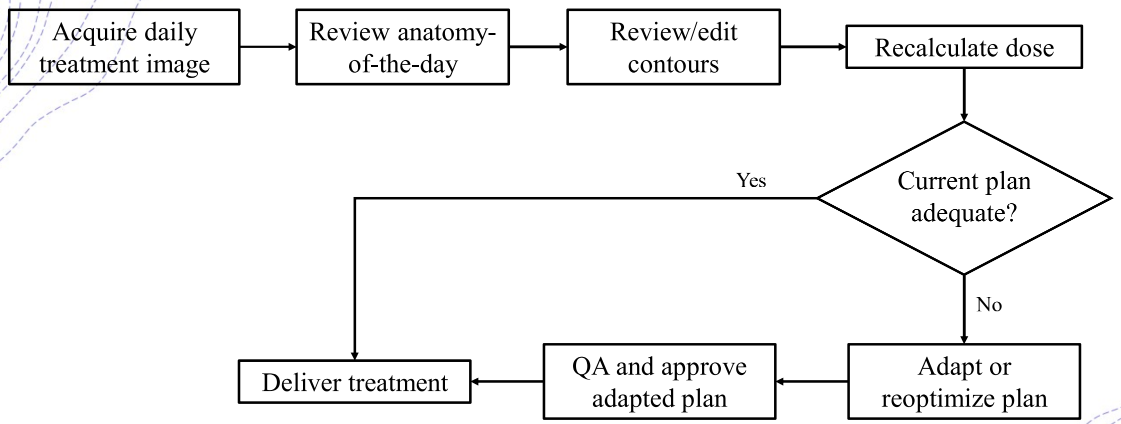
10

Online ART: Flowchart



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11

Online ART: Strengths and Limitations

Strengths
- Accounts for anatomy-of-the-day
- Can improve target coverage and OAR sparing
- Useful for daily organ filling or deformation
- Supports hypofractionation near critical structures
- Treatment is delivered in the same session

Limitations
- Increases treatment time each session
- Requires fast imaging, contouring, planning, and QA
- Creates time pressure for clinical decisions
- Requires coordinated team coverage
- Sensitive to intrafraction changes after adaptation

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12

Real-Time Adaptive Radiotherapy

- Real-time ART: adapts during beam delivery, rather than between fractions or before beam-on.
- Common triggers
  - Respiratory motion
  - Target drift during treatment
  - Organ motion during beam delivery
  - Patient motion after online adaptation
  - Motion-sensitive hypofractionated treatment
  - Tight target or OAR margins
  - …

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13

Real-Time Adaptive Radiotherapy

- Typical workflow
  1. Acquire continuous or repeated intrafraction imaging
  2. Monitor target and OAR motion during delivery
  3. Compare anatomy or target position with action threshold
  4. Adapt delivery in real time
  5. Continue, gate, track, or interrupt treatment
  6. Record delivery and imaging data for review

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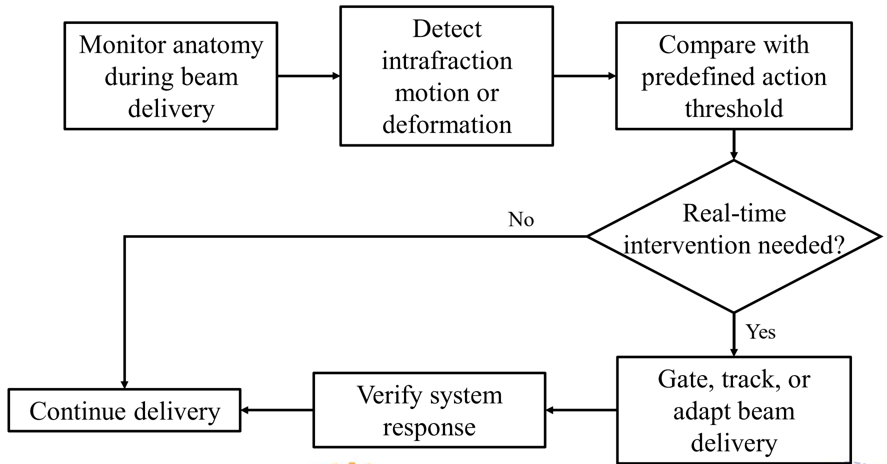
14

Real-Time ART: Flowchart



15

Real-Time ART: Strengths and Limitations

Strengths
- Addresses intrafraction motion during delivery
- Can reduce residual motion uncertainty
- Enables tighter margins in selected scenarios
- Useful for respiratory motion and target drift
- Supports highly conformal or hypofractionated treatments

Limitations
- Technically complex and platform-dependent
- Requires continuous or high-frequency imaging
- Sensitive to system latency and tracking accuracy
- QA is more challenging than offline or online ART
- Limited clinical implementation and evidence

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16

Comparison of ART Paradigms

Offline ART	Online ART	Real-Time ART
Mature	Growing rapidly	Emerging
Lower complexity	Moderate complexity	Highest complexity
Replanning between fractions	Daily adaptation	Continuous adaptation

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17

Key Take Aways

- Offline ART adapts between fractions
- Online ART adapts before delivery in the same session
- Real-time ART adapts during beam delivery
- Increasing responsiveness comes with increasing workflow and QA complexity
- The optimal approach depends on anatomy dynamics, disease site, technology, and resources

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18

Thank you!

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19